

Cross-Entropy

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First, an example

Before the formalism, a concrete picture. Let the true distribution be a fair coin, $p = (0.5, 0.5)$ on $\{H, T\}$. Shannon's entropy is $H(p) = 1$ bit per flip — the optimal average code length.

Now suppose we mistakenly design our code for the biased belief $q = (0.9, 0.1)$, assigning a short codeword to H and a long one to T . The average length we actually pay on data drawn from p is the cross-entropy

$$H(p, q) = -0.5 \log_2 0.9 - 0.5 \log_2 0.1 \approx 0.076 + 1.66 \approx 1.74 \text{ bits.}$$

Read aloud: “H of p comma q equals minus one half log base two of zero point nine minus one half log base two of zero point one, approximately one point seven four bits.” The cost of being wrong is the Kullback–Leibler divergence $D_{\text{KL}}(p \parallel q) = H(p, q) - H(p) \approx 0.74$ bits per symbol. This gap — extra bits paid for using the wrong model — is exactly what cross-entropy measures, and it is precisely the standard loss minimized in ML classification.

1 Definitions

Definition 1 (Discrete cross-entropy). Let p, q be probability mass functions on a countable alphabet $\mathcal{X}$, with $q(x) > 0$ whenever $p(x) > 0$. The *cross-entropy* of p with respect to q is

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = -\mathbb{E}_{X \sim p}[\log q(X)].$$

Read aloud: “H of p comma q equals minus the sum over x in script X of p of x times log q of x, which equals minus the expectation, X drawn from p, of log q of X.” We adopt $0 \log 0 = 0$. If there exists $x \in \mathcal{X}$ with $p(x) > 0$ and $q(x) = 0$, then $H(p, q) = +\infty$.

Definition 2 (Continuous cross-entropy). For probability densities f_p, f_q on $\mathbb{R}^d$ with $f_q > 0$ on the support of f_p ,

$$H(p, q) = - \int_{\mathbb{R}^d} f_p(x) \log f_q(x) dx.$$

Read aloud: “H of p comma q equals minus the integral over R-d of f-sub-p of x times log f-sub-q of x, d x.”

Definition 3 (Kullback–Leibler divergence). With the same hypotheses,

$$D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = H(p, q) - H(p),$$

Read aloud: “D-K-L of p given q equals the sum over x of p of x times log of p of x over q of x, which equals H of p comma q minus H of p.” where $H(p) = - \sum_x p(x) \log p(x)$ is Shannon's entropy.

2 Gibbs' inequality

Lemma 4 (Logarithm bound). *For all $z > 0$,*

$$\ln z \leq z - 1,$$

Read aloud: “natural log of z is less than or equal to z minus one.” with equality if and only if $z = 1$.

Proof. Let $f(z) = z - 1 - \ln z$ on $(0, \infty)$. Then $f'(z) = 1 - 1/z$, so $f'(z) < 0$ for $z \in (0, 1)$ and $f'(z) > 0$ for $z > 1$, with $f'(1) = 0$. Hence f has a strict global minimum at $z = 1$ with $f(1) = 0$. Therefore $f(z) \geq 0$ everywhere, with equality only at $z = 1$. $\square$

Theorem 5 (Gibbs' inequality). *For any two distributions p, q on a common alphabet $\mathcal{X}$ with q absolutely continuous with respect to p on the support of p ,*

$$H(p, q) \geq H(p),$$

Read aloud: “ H of p comma q is greater than or equal to H of p .” with equality if and only if $p(x) = q(x)$ for all x in the support of p .

Proof. We show the equivalent statement $D_{\text{KL}}(p \parallel q) \geq 0$. Restrict the sum to $\mathcal{X}_0 = \{x : p(x) > 0\}$; on this set $q(x) > 0$ by hypothesis, so $q(x)/p(x)$ is a positive real number.

Apply Lemma 4 with $z = q(x)/p(x)$ for each $x \in \mathcal{X}_0$:

$$\ln \frac{q(x)}{p(x)} \leq \frac{q(x)}{p(x)} - 1.$$

Read aloud: “natural log of q of x over p of x is less than or equal to q of x over p of x , minus one.” Multiplying both sides by $p(x) \geq 0$ preserves the inequality, and summing over $x \in \mathcal{X}_0$ gives

$$\sum_{x \in \mathcal{X}_0} p(x) \ln \frac{q(x)}{p(x)} \leq \sum_{x \in \mathcal{X}_0} (q(x) - p(x)) = \left(\sum_{x \in \mathcal{X}_0} q(x) \right) - 1 \leq 1 - 1 = 0,$$

Read aloud: “the sum over x in script- $\mathcal{X}$ -zero of p of x times log of q of x over p of x is at most the sum of q of x minus p of x , which equals the sum of q of x minus one, which is at most one minus one, equals zero.” where the last inequality uses $\sum_{x \in \mathcal{X}_0} q(x) \leq \sum_{x \in \mathcal{X}} q(x) = 1$. Converting to an arbitrary log base by dividing through by $\ln(\text{base})$ does not change the sign, so

$$-D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}_0} p(x) \log \frac{q(x)}{p(x)} \leq 0,$$

Read aloud: “minus D-K-L of p given q equals the sum over x in script- $\mathcal{X}$ -zero of p of x log of q of x over p of x , which is less than or equal to zero.” i.e. $D_{\text{KL}}(p \parallel q) \geq 0$. By the decomposition $H(p, q) = H(p) + D_{\text{KL}}(p \parallel q)$, this gives $H(p, q) \geq H(p)$.

Equality case. Suppose $H(p, q) = H(p)$, so $D_{\text{KL}}(p \parallel q) = 0$. Each summand $p(x) \ln(q(x)/p(x))$ is bounded above by $q(x) - p(x)$ (Lemma 4), and we must have equality in both the Lemma-4 step and the marginal step. Equality in Lemma 4 forces $q(x)/p(x) = 1$ for every $x \in \mathcal{X}_0$, i.e. $q(x) = p(x)$ on the support of p . The marginal step requires $\sum_{x \in \mathcal{X}_0} q(x) = 1$, which is now automatic since $\sum_{x \in \mathcal{X}_0} q(x) = \sum_{x \in \mathcal{X}_0} p(x) = 1$. Conversely, if $p = q$ on $\mathcal{X}_0$ then every term in D_{KL} vanishes, so $H(p, q) = H(p)$. $\square$

Corollary 6 (Decomposition). *For any p, q as above,*

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q),$$

Read aloud: “ H of p comma q equals H of p plus D - K - L of p given q .” and minimizing $H(p, q)$ over q (with p fixed) is equivalent to minimizing $D_{\text{KL}}(p \parallel q)$ over q .

Proof. Direct from the definitions: $-\sum_x p(x) \log q(x) = -\sum_x p(x) \log p(x) + \sum_x p(x) \log \frac{p(x)}{q(x)}$. The second statement follows because $H(p)$ does not depend on q . $\square$

Example 7 (Fair coin, biased code). Let $p = (\frac{1}{2}, \frac{1}{2})$ and $q = (0.9, 0.1)$ on $\mathcal{X} = \{0, 1\}$. Then $H(p) = 1$ bit, and

$$H(p, q) = -\frac{1}{2} \log_2 0.9 - \frac{1}{2} \log_2 0.1 \approx 1.737 \text{ bits},$$

Read aloud: “ H of p comma q equals minus one half log base two of zero point nine minus one half log base two of zero point one, approximately one point seven three seven bits.” so $D_{\text{KL}}(p \parallel q) \approx 0.737$ bits — the per-symbol coding overhead of using a code matched to q on data drawn from p .